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AI Engineer Roadmap 2027

4 simple phases. Real projects. No random tutorials.

The roadmap at a glance

PHASE 1	**Python + Maths + Data tools**	Python basics, linear algebra, calculus, statistics, probability, NumPy and Pandas
PHASE 2	**Core Machine Learning**	Build core models yourself with NumPy, learn evaluation techniques, practise with Pandas and libraries
PHASE 3	**Deep Learning**	Neural networks, CNN, RNN, Transformers, transfer learning with PyTorch and TensorFlow
PHASE 4	**Ship it like an engineer**	Version control, containerization, MLOps tools

How to use this PDF

- Go phase by phase. Do not jump ahead.
- Every phase has a checklist, free links, and project ideas. Build the projects, do not only watch videos.
- From Phase 1 itself, build one habit: read case studies and AI papers. See what works, what does not, and why it fails.
- Put every project on GitHub. It becomes your portfolio.

Note: free resources are marked in the link tables. Websites change, so if a link moves, search the resource name.

PHASE 1

Python + Maths + Data tools

Foundation. Make this solid.

What to learn

- [ ] **Python basics:** variables, functions, coding exercises, data structures
- [ ] **Maths:** linear algebra, calculus, statistics, probability (spend a little extra time here)
- [ ] **NumPy and Pandas:** data manipulation, visualization and analysis
- [ ] **Habit from day one:** read case studies and AI papers

Where to learn (free)

Topic	Resource	Link
Python basics	Kaggle Learn: Python (hands-on, in browser)	www.kaggle.com/learn/python
Python basics	CS50P: Harvard's intro to Python	cs50.harvard.edu/python
Python reference	Official Python Tutorial	docs.python.org/3/tutorial
Data structures	Problem Solving with Algorithms and Data Structures using Python	runestone.academy/ns/books/published/pswadsup/pythonds-3.html
Linear algebra	Khan Academy: Linear Algebra	www.khanacademy.org/math/linear-algebra
Linear algebra (visual)	3Blue1Brown: Essence of Linear Algebra	www.3blue1brown.com/topics/linear-algebra
Calculus	3Blue1Brown: Essence of Calculus	www.3blue1brown.com/topics/calculus
Stats + probability	Khan Academy: Statistics and Probability	www.khanacademy.org/math/statistics-probability
Maths for ML	Mathematics for Machine Learning (free book PDF)	mml-book.com
NumPy	NumPy: the absolute basics for beginners	numpy.org/doc/stable/user/absolute_beginners.html
Pandas	10 minutes to pandas	pandas.pydata.org/docs/user_guide/10min.html
Pandas	Kaggle Learn: Pandas	www.kaggle.com/learn/pandas
Reading papers	arXiv paper example: Attention Is All You Need	arxiv.org/abs/1706.03762
Reading papers	Hugging Face Papers (trending AI papers)	huggingface.co/papers

Project examples

- **Expense tracker (Python)** - add expenses, save them to a file, show the monthly total. Uses variables, functions and data structures.
- **Maths in code (NumPy)** - write your own functions for mean, variance, dot product and matrix multiplication, then check them against NumPy.
- **Data detective (Pandas)** - pick any CSV (for example the Titanic dataset from kaggle.com/datasets), find 5 insights and make 3 charts.
- **Paper notes** - read one case study or AI paper and write 3 lines: what worked, what did not, why it failed.

> **Move on when:** you can write small Python programs without looking up basic syntax, and you can load a CSV, clean it and chart it with Pandas.

PHASE 2

Core Machine Learning

Understand models by building them.

What to learn

- [ ] **Core ML models:** understand how different types of models are created
- [ ] **Build them yourself** with NumPy, so you understand the architecture and do not only see it from the outside
- [ ] **Evaluation techniques:** learn how each kind of model is evaluated
- [ ] **Practice:** create models with Pandas and the relevant libraries

Where to learn

Topic	Resource	Link
ML basics (free)	Kaggle Learn: Intro to Machine Learning	www.kaggle.com/learn/intro-to-machine-learning
ML course (Coursera)	Machine Learning Specialization by Andrew Ng (check audit option; certificate is paid)	www.coursera.org/specializations/machine-learning-introduction
Evaluation (free)	scikit-learn: metrics and scoring	scikit-learn.org/stable/modules/model_evaluation.html
Maths behind models (free)	Mathematics for Machine Learning, Part II (regression, PCA, SVM)	mml-book.com
Practice data (free)	Kaggle Datasets	www.kaggle.com/datasets

Project examples

- **Linear regression from scratch** - predict house price from area using only NumPy. Write the cost function and gradient descent yourself.
- **kNN or logistic regression from scratch** - build one more model in NumPy on a small dataset (for example, Iris flowers).
- **Spam or survival classifier** - use Pandas plus scikit-learn to train 2 or 3 models on the same data (for example, Titanic survival).
- **Evaluation report** - compare those models with accuracy, precision, recall and a confusion matrix. Write which model you pick and why.

> **Move on when:** you can explain how a model learns, you have built at least two models from scratch, and you can judge a model with the right evaluation metric.

PHASE 3

Deep Learning

The most important phase. Build strong projects here.

What to learn

- [ ] **Neural network fundamentals**
- [ ] **CNN** (images), **RNN** (sequences), **Transformers**, **transfer learning**
- [ ] **Build everything with PyTorch and TensorFlow**
- [ ] **Build strong projects.** This is where you see why Phase 4 is needed.

Where to learn

Topic	Resource	Link
PyTorch (free)	PyTorch: Learn the Basics	pytorch.org/tutorials/beginner/basics/intro.html
TensorFlow (free)	TensorFlow 2 quickstart for beginners	www.tensorflow.org/tutorials/quickstart/beginner
Neural nets from scratch (free)	Andrej Karpathy: Neural Networks, Zero to Hero	karpathy.ai/zero-to-hero.html
Deep learning course (Coursera)	Deep Learning Specialization by Andrew Ng (check audit option; certificate is paid)	www.coursera.org/specializations/deep-learning
Transformers paper	Attention Is All You Need	arxiv.org/abs/1706.03762
Keep reading papers	Hugging Face Papers	huggingface.co/papers

Project examples

- **Digit recogniser (neural network)** - train a network on the MNIST handwritten digits dataset with PyTorch.
- **Image classifier (CNN)** - classify images such as cats vs dogs, or CIFAR-10 classes. Then rebuild it in TensorFlow.
- **Text model (RNN)** - classify movie review sentiment, or generate text one character at a time.
- **Transfer learning project** - take a pretrained model and fine-tune it on your own small dataset (for example, sorting customer messages into topics).
- **Portfolio rule** - finish 2 or 3 strong projects with a clear README: problem, data, model, result, what failed.

> **Move on when:** you have built a CNN, an RNN and a transfer learning project, and you can show results plus what did not work.

PHASE 4

Ship it like an engineer

Version control, containerization, MLOps tools.

What to learn

- [ ] **Version control** (Git and GitHub)
- [ ] **Containerization** (Docker)
- [ ] **MLOps tools** to run, track and repeat your ML work

Where to learn (free)

Topic	Resource	Link
Version control	Pro Git book	git-scm.com/book/en/v2
Containers	Docker: Get started	docs.docker.com/get-started
MLOps	Made With ML: MLOps course	madewithml.com

Project examples

- **GitHub portfolio** - push every project from Phases 1 to 3 to GitHub with small, clear commits and a README.
- **Containerize a model** - wrap your best model in a small API and run it inside a Docker container.
- **MLOps pass** - follow Made With ML and apply its practices (versioning, testing, reproducibility) to one of your projects.
- **Final capstone** - one project end to end: data, model, evaluation, API, container, GitHub README. This is the project you show in interviews.

> **Move on when:** your best project runs on someone else's machine with one command, and your GitHub profile tells the full story.

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youtube.com/channel/UCzn888DYJfy2AfbrOkduhZg

Don't be a typist. Be an Architect.